Reverse Transcription

M10 | ver 1.1 | Revised 2026-10-03

SUSTech-BIO 2026 iGEM Team

Purpose

Prepare cDNA from 850 ng total RNA using the Takara PrimeScript RT reagent Kit with gDNA Eraser (Perfect Real Time), RR047A, with the TB Green recipe. [1]

Materials & Equipment

Item	Supplier / identifier	Use
PrimeScript RT reagent Kit with gDNA Eraser (Perfect Real Time)	Takara / RR047A	DNA removal and cDNA synthesis
Purified total RNA	Sample-specific	Measured concentration; assess integrity
RNase-free tubes and filter tips	Axygen tips; laboratory supplies	Reaction setup
Pipettes, ice rack and microcentrifuge	Eppendorf centrifuge; laboratory pipettes and ice rack	Dispensing and collection
Thermal cycler	Same laboratory thermal cycler as M01	42°C, 37°C and 85°C stages

Before You Start

Use 850 ng total RNA per reaction. This team-selected input is within the manufacturer's 1 µg limit for the TB Green recipe. [1]

Calculate RNA volume V (µL) = 850 / RNA concentration (ng/µL). V must be ≤7 µL, requiring RNA at ≥121.43 ng/µL. If RNA is more dilute, concentrate or re-extract it and remeasure before setup; do not exceed the 10 µL DNA-removal volume.

Keep kit stocks at -20°C. Vortex both buffers and briefly spin; collect viscous enzymes by a short spin and pipette slowly. Assemble reactions on ice. [1]

Include a matched no-RT control where DNA could amplify: process the same RNA through DNA removal, then replace the 1 µL RT enzyme with water. Include a no-template control in the downstream PCR.

Procedure

A. Remove genomic DNA

1. Combine the following for each sample. Use each sample's measured RNA concentration to calculate its RNA and water volumes. [1]

Component	Volume per reaction
5× gDNA Eraser Buffer	2 µL
gDNA Eraser	1 µL
Total RNA (850 ng)	V = 850 / concentration (ng/µL)
RNase-free water	(7 - V) µL
Total	10 µL

2. Incubate at 42°C for 2 min, then place at 4°C while preparing the next stage. [1]

B. Synthesize cDNA

3. Prepare 10 µL of RT addition mix per sample, excluding the treated RNA. Add it to the entire 10 µL DNA-removal reaction, mix gently and start the program promptly. [1]

Component	Volume
DNA-removal reaction	10 µL
5× PrimeScript Buffer 2 (for Real Time)	4 µL
PrimeScript RT Enzyme Mix I	1 µL
RT Primer Mix	1 µL
RNase-free water	4 µL
Final volume	20 µL

4. Run 37°C for 15 min, followed by 85°C for 5 sec and a 4°C hold. [1]

C. Store and use cDNA

5. Store products at or below -20°C for longer-term use. For qPCR, keep carried-over RT reaction at ≤10% of the PCR volume; this corresponds to ≤2 µL undiluted product in 20 µL PCR. Record any dilution. [1]

p.s.

Retain stage A even after upstream DNase treatment: its buffer contributes to the complete RT system. [1]

For individual additions in stage B, mix water and Buffer 2 with the treated RNA before adding primers and RT enzyme. This suppresses residual Eraser activity first. [1]

Read the final heating step in seconds: 85°C for 5 sec. Keep the 37°C RT incubation at 15 min. [1]

When processing samples of different RNA concentrations, calculate water separately for each tube. Normalize RNA concentration first if using one shared DNA-removal master mix.

Label no-RT tubes before setup and dispense their enzyme-free mix separately. Carry controls through the same downstream dilution and assay as their matching samples.

Expected Outcome

Obtain 20 µL of cDNA reaction mixture per sample. Evaluate it with a validated downstream assay and its controls; tube appearance does not establish success. A positive no-RT result indicates possible DNA contribution, and a positive no-template result requires assessment of contamination or assay background before interpreting RNA-derived signal.

Troubleshooting

Observation	Check first	Action
No or weak downstream signal	RNA integrity, input calculation, enzyme addition and recorded program.	Use a known amplifiable control to distinguish assay failure from sample failure. Re-extract compromised RNA or correct the identified setup error.
Signal in the no-RT control	DNA-removal setup and contamination during handling.	Check Eraser addition and the 42°C stage. Resolve DNA carryover before attributing amplification to RNA.
Variable signal across comparable samples	RNA input, dilution records and dispensing consistency.	Reconcile calculations and compare controls. Repeat affected preparations with consistent inputs and a shared mix.
Signal improves unexpectedly after dilution	RT carryover and extraction contaminants.	Check the ≤10% carryover limit. Compare matched dilutions with controls; clean up or re-extract RNA if inhibition persists.

Reference

[1] Takara Bio. PrimeScript RT reagent Kit with gDNA Eraser (Perfect Real Time), RR047A. Product Manual, v202202Da, pp. 3-7. Accessed 2026-09-27. Open source